OBINNA DOMINIC UKE

Applied Environmental Economist ◇ Climate, Energy, and Finance

dominic.uke@um6p.ma ◇ ukeobinna@gmail.com ◇ +212 623 755 857

obinnaud.github.io ◇ linkedin.com/in/obinnauke

Africa Business School, Mohammed VI Polytechnic University (UM6P), Rabat, Morocco
COPPE, Federal University of Rio de Janeiro (UFRJ), Rio de Janeiro, Brazil

FIELDS

Primary: Environmental and Energy Economics
Secondary: Financial Economics, Development Economics, Applied Econometrics

Research interests: Environmental and energy economics; carbon pricing and fossil fuel subsidies; carbon border adjustment and transition risk; climate risk and financial markets; global value chains and structural transformation; sovereign risk and credit markets; climate risk transmission and macroeconomic dynamics; climate finance and adaptation in the Global South.

EDUCATION

Ph.D. in Management and Organizational Sciences 2023 to Aug. 2027 (expected)

Research field: Environmental and Energy Economics
Africa Business School, Mohammed VI Polytechnic University (UM6P), Rabat, Morocco

- Dissertation: *Climate Risk, Carbon Pricing, and Financial Markets: Essays in Environmental and Financial Economics*
- Supervisors: Prof. Amine Ben Amar (Finance Cluster, ABS, UM6P) and Prof. Olga Gorelkina (Moroccan Center for Game Theory, UM6P)

Joint Doctorate, Production Engineering (PEP/COPPE) 2023 to Aug. 2027 (expected)

Research field: Environmental and Energy Economics
COPPE, Federal University of Rio de Janeiro (UFRJ), Rio de Janeiro, Brazil

- Joint doctoral agreement with ABS, UM6P, conferring the doctoral degrees of both institutions on a single defence; co-supervisor Prof. Luan Santos

M.Sc. in Fertilizer Science and Technology Oct. 2020 to Jul. 2022
ESAFE, Mohammed VI Polytechnic University (UM6P), Benguerir, Morocco

B.Sc. in Agriculture (Soil and Crop Sciences) Dec. 2013 to Aug. 2018
Usmanu Danfodiyo University, Sokoto, Nigeria

JOB MARKET PAPER

"Fossil Fuel Underpricing Blocks Disaster-Induced Structural Transformation: Evidence from Carbon-Intensive Global Value Chains"

Sole authored. Under review: Structural Change and Economic Dynamics. JEL: Q54, Q58, L16, O14, H23, F18.

Using a country-sector-year panel of 77 countries, 18 manufacturing sectors and seven years (2016 to 2022) under triple fixed effects (country-sector, country-year, sector-year), I show that fossil fuel underpricing blocks the reallocation away from carbon-intensive sectors that climate disasters would otherwise induce. The triple-interaction coefficient is 2.63 ($p < 0.01$). The marginal effect of a climate disaster on carbon-intensive global value chain output flips sign at an underpricing level of 1.845% of GDP, the 68.5th percentile of the country distribution, and 31.5% of country-sector-year

cells sit on the lock-in side of that threshold. Lock-in concentrates in fossil-fuel-exporting, non-high-income economies with weaker governance and higher resource dependence. The result holds in all 77 leave-one-country-out estimates and survives a horse race against sanctions and conflict channels.

WORKING PAPERS

"Riding the Climate Wave: A Wavelet Analysis of Sovereign Risk in Developed and Developing Economies"

With Amine Ben Amar and Olga Gorelkina. Revise and resubmit: International Review of Financial Analysis (FINANA-D-25-05094). Presented: 2nd QFDAM, Amiens, June 2025; 71st EWGCFM, Rabat, May 2025.

We ask which dimension of climate risk sovereign credit markets price: physical exposure (vulnerability) or the capacity to adapt (readiness). Using 59 economies over 2008 to 2021, spanning 28 advanced economies and 31 emerging market and developing economies, wavelet coherence documents significant co-movement between climate vulnerability and sovereign credit default swap spreads at short, medium and low frequencies, neither concentrated in the long horizon nor systematically different between advanced economies and EMDEs. In the cross-section the vulnerability gradient vanishes once development is controlled, while readiness carries a strong negative association with spreads net of income, public debt and the Worldwide Governance Indicators: for EMDEs, closing the readiness disadvantage relative to advanced economies is associated with roughly 74% smaller spreads. It is adaptive capacity, not exposure, that the market rewards. We also document and correct a measurement convention that inflates conventional wavelet coherence summaries.

"Climate Shocks and Food Manufacturing: Propagation, Persistence, and Structural Change in the United Kingdom"

With Amine Ben Amar and Olga Gorelkina. Submitted: Economic Modelling. Presented: 3rd QFDAM, Rabat, June 2026. JEL: C32, Q54, L66.

Using UK quarterly data for 1990 to 2025, we separate primary agriculture from downstream food manufacturing and combine a VARX baseline with time-varying local projections with stochastic volatility. A one standard deviation temperature shock raises agricultural output by about 1.1%, while food-manufacturing estimates turn negative, peaking near -0.24% and remaining negative for three years, though not individually significant at conventional levels. The opposite-sign profile holds under energy-price controls and alternative specifications. The time-varying estimates show food manufacturing's domestic climate sensitivity weakening from the late 2010s, consistent with UK food-import dependence rising from 38% to over 50%, while agriculture's response strengthens at short horizons. Measured domestic temperature sensitivity has shifted upstream to farming.

"Fiscal Capacity, Not Climate Exposure: Decomposing Global Fossil-Fuel Underpricing and Its Equity Burden"

With Olga Gorelkina and Amine Ben Amar. Submitted: Energy Policy. JEL: Q48, Q54, Q58, H23, H20, O23.

We construct a near-global panel of effective carbon prices for 174 countries from 2015 to 2022, integrating explicit carbon-pricing instruments with implicit fossil fuel subsidies against three social cost of carbon benchmarks, and decompose the underpricing shortfall into a domestic wedge, made up of supply-cost gaps, foregone consumption taxes and local air pollution, and a global climate wedge. Underpricing is near-universal: only Sweden, Norway and Finland price above zero. Fiscal capacity, measured by non-resource tax revenue, predicts the domestic wedge, which is roughly two thirds of the total, while no observable state characteristic predicts the climate wedge. Climate vulnerability does not raise deficits directly, yet vulnerability and weak fiscal capacity coincide so closely that the income-normalised burden is about twenty times larger in the most vulnerable quintile. Domestic revenue mobilisation is therefore climate policy for the larger part of the shortfall, while the climate wedge requires international coordination.

"Financial Reach and Household Coping under Climate Shocks: Evidence from Ten

African Countries"

With Todd Cort (Yale School of Management). Submitted: World Development. JEL: Q54, O16, G51, D14.

Harmonised shock-and-coping modules from ten African countries permit a within-household comparison of responses to a covariate climate shock and to an idiosyncratic one, so household-fixed characteristics difference out. Covariate shocks are associated with a 1.9 to 2.2 percentage-point higher probability of reducing consumption, while idiosyncratic shocks more often trigger asset sales. In the 2018 West African sample a sender elsewhere in the country is associated with a 4.6 percentage-point smaller shift towards cutting consumption than a same-region sender, surviving administrative-region fixed effects, while credit access is statistically indistinguishable from zero. Growing-season rainfall anomalies remain correlated across roughly 254 km, measured from a CHIRPS correlogram, which motivates but does not identify the reach measure. The estimates are associational. The contribution is conceptual as much as empirical: instrument ownership alone is an incomplete measure of climate resilience, because the exposure of the income backing a financial claim matters alongside the instrument a household holds.

"Building Resilience, Reducing Risk: A Systematic Review of Climate Resilience, Climate Finance Impacts, and Market Mispricing in the Global South"

With Amine Ben Amar and Olga Gorelkina. Retained as a dissertation chapter.

A PRISMA-based systematic literature review consolidating the relationship between climate resilience initiatives, financial risk reduction and market mispricing in the Global South. Firms in climate-vulnerable regions face higher borrowing costs and banks impose stricter loan terms in high-risk geographies. Effective governance and precautionary financial policies are identified as critical moderators of climate finance impact.

WORK IN PROGRESS

"External Carbon Pricing and Transition Risk in Emerging Markets: Evidence from African Exporters under the EU Carbon Border Adjustment Mechanism"

Target: Business Strategy and the Environment.

I build an exporter-by-tariff-line measure of CBAM exposure for African economies from the Commission's country-specific default values, Eurostat Comext trade at CN8 and the quarterly certificate price, using longest-prefix concordance because defaults are published at mixed granularity. On 2025 trade the chargeable liability is EUR 45 million in 2026 under the phase-in schedule and EUR 1.80 billion from full phase-in in 2034, against a 2023 to 2025 mean of EUR 1.38 billion. Two findings drive the paper. First, the ad valorem equivalent rises with processing stage in all four covered sectors while EU applied tariffs are flat and low, reaching 48.2% on semi-finished steel against a 0.00% most-favoured-nation tariff, so the mechanism reintroduces the tariff escalation that trade policy removed. Second, exposure resolves to individual installations, with one production line accounting for between 88% and 98% of national liability in several countries, which makes sovereign and single-firm transition risk the same object. Because CBAM credits the carbon price explicitly paid rather than the price net of fossil subsidies, the distance to reform is wider than the regulation's own arithmetic implies.

"Does Support Arrive When the Water Does? Financial Adaptation Capacity of Nigerian Non-Farm Enterprises"

Sole investigator.

Carries the covariance logic from households to firms. Event study of continuous flood exposure across five waves of the Nigeria General Household Survey Panel and the high-frequency NLPS rounds around the 2022 floods, combining NIHSA flood outlooks, Sentinel-1, CHIRPS and ERA5, together with a planned enterprise survey in the Anambra River basin.

"Sovereign Climate Risk and Corporate Bond Spreads: Evidence from Emerging Markets"

With Luan Santos (COPPE, UFRJ).

Panel of 30 emerging markets and roughly 8,000 corporate bonds from 2018 to 2024; fixed effects and instrumental variables; implications for the IFRS S1 and S2 climate disclosure standards. Developed under the joint doctoral agreement with COPPE, UFRJ.

PEER-REVIEWED PUBLICATIONS

Guéablé, Y.K.D., **Uke, O.D.** et al. (2022). Study of the Sodicity of Phosphate By-Products and Sludge Mixture for Large-Scale Application in Mine Site Reclamation. *Environmental Sciences Proceedings*, 16(1), 18.

Uke, O.D. & Haliru, M. (2021). Salinity Study of the Soils of Fadama Farms, Sokoto, Nigeria. *Farming and Management*, 6, 1-7.

Yusuf, Y.M., Bagudo, H.A. & **Uke, O.D.** (2021). Effects of Weed Control on Growth and Yield of Pearl Millet in the Sudan Savanna, Sokoto State, Nigeria. *Journal of Current Opinion in Crop Science*, 2(2), 155-164.

CONFERENCE PRESENTATIONS

3rd Quantitative Finance and Data Analytics Meeting (QFDAM 2026), Rabat, Morocco June 2026
Climate Shocks, Sectoral Transmission, and Monetary Policy: Evidence from UK Food Manufacturing

2nd Quantitative Finance and Data Analytics Meeting (QFDAM 2025), Amiens, France June 2025
Riding the Climate Wave: Wavelet Analysis of Sovereign Risk in Developed and Developing Economies

71st EURO Working Group for Commodities and Financial Modelling (EWGCFM 2025), Rabat, Morocco May 2025
Riding the Climate Wave: Wavelet Analysis of Sovereign Risk in Developed and Developing Economies

TEACHING

Teaching Assistant, Financial Econometrics [Master 2] Sep. to Nov. 2026
Mohammed VI Polytechnic University (UM6P) (Instructor: Prof. Amine Ben Amar)

Teaching Assistant, Microeconomics [Licence 1] Sep. to Nov. 2026
Mohammed VI Polytechnic University (UM6P) (Instructor: Prof. Amine Ben Amar)

Teaching Assistant, Essentials of Financial Markets [M1] Jan. to Feb. 2025
Africa Business School, UM6P (Instructor: Prof. Amine Ben Amar)

PROFESSIONAL EXPERIENCE

PhD Researcher, Environmental and Energy Economics Oct. 2023 to Present
Africa Business School, UM6P, Morocco; COPPE, UFRJ, Brazil

Produce policy-relevant research on climate shock propagation, physical risk pricing, carbon pricing failures and sovereign debt vulnerability. Build and validate multi-source climate, trade, firm and household datasets, and publish reproducible replication packages.

Regional Field Project Manager Jul. 2022 to Oct. 2023
OCP Africa Nigeria

Coordinated multi-regional agricultural development programmes across Southeast and South-South Nigeria in partnership with USAID/TradeHub. Managed programme budgets, donor reporting and workplan tracking. Produced policy briefs on climate-smart agriculture for government counterparts and programme leadership.

Research Agronomist / Blending Plant Support Jun. 2021 to Jun. 2022
OCP Africa Nigeria

Designed and executed fertilizer validation field trials; prepared grant applications and supported monitoring and evaluation for USAID-funded agricultural initiatives.

Founder and Chief Technical Officer Dec. 2020 to Jan. 2022
AgriBox SARL, Morocco

Founded a climate-smart hydroponic enterprise; developed impact investment proposals and private sector partnerships.

Project Manager, Microfinance Operations Jun. 2019 to Mar. 2020
Bincom ICT Solutions (Social Lender), Nigeria

Managed microfinance loan disbursement, risk assessment and financial proposals across Kano State. Served as institutional liaison between Social Lender and the banking sector.

FELLOWSHIPS AND AWARDS

Full Doctoral Merit Scholarship, Africa Business School, UM6P, Morocco	2023
HULT Prize in Innovation, Mohammed VI Polytechnic University	2021
NARSAD Deliberative Polling Award, UM6P and Stanford University	2021
IBN Rochid Foundation Scholarship, M.Sc. Studies, OCP Group	2020

TECHNICAL SKILLS AND LANGUAGES

Statistical software: R, Stata, Python, SQL, GIS, VS Code, Git, Advanced Excel.

Methods: SVAR and local projections, including time-varying local projections with stochastic volatility; wavelet and partial wavelet coherence; triple fixed effects with cluster-robust inference; within-household first differences; event study designs with continuous exposure; IV and weak-instrument diagnostics; leave-one-out robustness; PRISMA systematic review.

Data and geospatial: construction and validation of multi-source country, sector, firm and household panels; CHIRPS, CHIRTS, ERA5 and Sentinel-1 products; Eurostat Comext and UN Comtrade at tariff-line level; World Bank LSMS-ISA and EHCVM household surveys; IMF fossil fuel subsidy and carbon pricing data; reproducible pipelines with replication packages deposited on Zenodo.

Languages: English (native), Igbo (native), Portuguese (intermediate), French (basic).